

PROJECT DOCUMENTATION

Digital Lending: Portfolio Optimization

Consulting & Analytics Club · IIT Guwahati · June 2026

Project Title	Digital Lending: Portfolio Optimization
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1. Project Overview

This project was developed as part of the Consulting & Analytics Club (CA Club), IIT Guwahati's project programme on Digital Lending: Portfolio Optimization. The objective was to design a data-driven credit policy framework for a digital lending institution — covering risk segmentation, creditworthiness evaluation, early warning signals, and portfolio optimization strategy.

The project follows a full end-to-end analytical pipeline: synthetic dataset generation → exploratory data analysis → predictive modelling → SHAP feature attribution → CLV revenue analysis → credit policy design → impact quantification.

1.1 Problem Statement

How can a digital lending institution leverage end-to-end customer and transaction data to strengthen credit risk assessment, enable early detection of delinquencies, optimise pricing and product strategies, and drive sustainable, risk-adjusted portfolio growth?

1.2 Five Key Questions Addressed

#	Key Question
Q1	Which customer segments show materially different risk and repayment behaviours?
Q2	How do acquisition channels impact portfolio quality and customer lifetime value?
Q3	Which loan products, ticket sizes, and tenures deliver the best balance of growth and risk?
Q4	How can pricing and tenure strategies be tailored to improve overall portfolio outcomes?
Q5	What metrics should senior leadership monitor to proactively manage risk and growth?

2. Submission File Structure

The final submission is a single ZIP file containing the following deliverables:

File	Description
■ Credit_Policy_Report_Final.pdf	Main deliverable. 7-section credit policy report addressed to the Chief Risk Officer. Covers risk segmentation, CLV analysis, early warning framework, credit policy recommendations, and projected impact assessment.
■ Digital_Lending_Presentation.pptx	10-slide presentation deck covering all key findings, charts, and policy recommendations. Designed for stakeholder presentation.
■ Digital_Lending_Notebook.ipynb	Google Colab notebook containing all Python code — synthetic dataset generation, data cleaning, EDA, feature engineering, Logistic Regression, XGBoost, SHAP analysis, CLV computation, and all chart generation.
■ data/digital_lending_synthetic.csv	10,000-row synthetic dataset generated per project specifications. Contains 21 features across 7 data areas: customer profile, loan & product, repayment behaviour, behavioural signals, acquisition channel, time dimension, outcomes.
■ README_Documentation.pdf	This file. Technical documentation covering project overview, file structure, how to run the notebook, dataset description, methodology, and key results.
■ Dashboard.html	Interactive single-file HTML dashboard. Visualises default rates by grade, CLV analysis, channel performance, and early warning KPIs. Open in any browser.

3. How to Run the Notebook

3.1 Environment

The notebook was developed and tested on **Google Colab** (free tier). No local installation required. All dependencies are installed within the notebook.

3.2 Step-by-Step Instructions

Step	Action	Detail
Step 1	Open Google Colab	Go to colab.research.google.com → Click "Upload" → Select Digital_Lending_Notebook.ipynb
Step 2	Run All Cells in Order	Click Runtime → Run All. Or run each cell individually top to bottom. Do not skip cells — each cell depends on the previous.
Step 3	Dataset Generation	Cell 1 generates the synthetic dataset automatically. No external download needed. Output: digital_lending_synthetic.csv saved to Colab workspace.
Step 4	Review Outputs	Each section prints summary statistics and saves charts as .png files. All charts are also displayed inline below each cell.
Step 5	Download Files	Run the final download cell to save all outputs to your computer.

3.3 Required Libraries

All libraries below are pre-installed on Google Colab. If running locally, install with: `pip install [library name]`

Library	Purpose
pandas	Data manipulation and analysis
numpy	Numerical computations
matplotlib	Chart generation
seaborn	Statistical visualisations
scikit-learn	Logistic Regression, train/test split, metrics
xgboost	XGBoost classifier
shap	SHAP feature importance analysis

4. Dataset Description

The dataset was synthetically generated per project specifications, which required participants to design and create their own dataset reflecting a realistic digital lending ecosystem. The dataset is not sourced from Kaggle or any external repository — it was programmatically generated using numpy with controlled random seeds to ensure reproducibility.

4.1 Dataset Specifications

Specification	Value
Total Rows	10,000 loan records
Total Features	21 columns
Target Variable	default (1 = defaulted, 0 = fully repaid)

Specification	Value
Overall Default Rate	42.5%
NTC Customers	3,488 (34.88% — customers with no credit history)
Random Seed	42 (fully reproducible)
File Format	CSV (digital_lending_synthetic.csv)

4.2 Feature Descriptions

Feature	Description
customer_id	Unique identifier (CUST00001 format)
age	Borrower age (21–57 years)
city_tier	City tier (1=Metro, 2=Tier-2, 3=Semi-urban)
employment_type	Salaried / Self-Employed / Informal / Student
monthly_income	Monthly income in INR
has_credit_history	1 = has bureau history, 0 = NTC customer
fico_score	Credit bureau score (580–850; NaN for NTC)
acquisition_channel	App Organic / Referral / Agent / Partnership
loan_purpose	Personal / BNPL / Working Capital / Education / Medical
loan_amount	Loan amount in INR (5,000–500,000)
tenure_months	Loan tenure (3/6/12/18/24/36 months)
interest_rate	Annual interest rate (%)
monthly_installment	Calculated monthly EMI in INR
grade	Risk grade A–G (derived from FICO + income)
dti	Debt-to-income ratio
installment_to_income	Monthly EMI as fraction of monthly income
cash_flow_volatility	Income volatility score (0–1)
spending_shock	1 = recent spending spike detected, 0 = normal
days_delinquent	Days past due (0 for non-defaulters)
partial_payment_ratio	Fraction of loan repaid (0–1)
default	Target variable: 1 = defaulted, 0 = fully repaid

4.3 Data Design Areas

The dataset was designed to cover all 7 data areas specified in the project brief:

Data Area	Features Covering This Area
Customer Profile	age, city_tier, employment_type, monthly_income, has_credit_history, fico_score

Data Area	Features Covering This Area
Loan & Product	loan_purpose, loan_amount, tenure_months, interest_rate, monthly_installment, grade
Repayment Behaviour	days_delinquent, partial_payment_ratio, default
Behavioural Signals	cash_flow_volatility, spending_shock
Acquisition	acquisition_channel
Time Dimension	tenure_months (origination and repayment period proxy)
Outcomes	default, partial_payment_ratio (value proxy)

5. Methodology

5.1 Analytical Pipeline

#	Step	Description
1	Synthetic Data Generation	10,000-loan dataset generated with numpy using controlled default probability functions incorporating grade, DTI, employment type, channel, and behavioural signals.
2	Exploratory Data Analysis	6 analytical charts: default rate by grade, channel, employment type, loan size within grade, CLV vs risk scatter, expected revenue by grade.
3	Gap Analysis	Mean comparison of 6 key metrics (DTI, FICO, installment/income, income, loan amount, interest rate) between defaulters and non-defaulters.
4	Feature Engineering	Derived features: installment_to_income ratio, grade_num (A=1 to G=7), channel_num, emp_num encodings for modelling.
5	Predictive Modelling	Logistic Regression (AUC: 0.699) and XGBoost (AUC: 0.693) trained on 80/20 train-test split with class_weight balancing for default imbalance.
6	SHAP Analysis	TreeExplainer applied to XGBoost on 1,000 test samples to identify and rank top default predictors by mean absolute SHAP value.
7	CLV Revenue Analysis	CLV proxy = loan_amount × interest_rate/100 × tenure_months/12 computed for non-defaulters. Expected revenue = CLV × (1 - default_rate) per grade.
8	Credit Policy Design	Grade-based approval framework, employment rules, channel rules, behavioural guardrails, and NTC three-gate evaluation framework derived from data findings.

5.2 Model Details

Parameter	Logistic Regression	XGBoost
AUC-ROC	0.6985	0.6933
Train/Test Split	80% / 20%	80% / 20%
Class Balancing	class_weight='balanced'	scale_pos_weight=2
Key Parameters	max_iter=1000	n_estimators=100, max_depth=4, lr=0.1
Use in Report	Interpretable policy levers	SHAP feature attribution

6. Key Results Summary

6.1 Critical Numbers

Metric	Value
Overall Default Rate	42.5%
Grade A Default Rate	14.0%
Grade G Default Rate	62.1% (4.4x Grade A)
Grade C Expected Revenue/Loan	Rs. 14,154 (highest risk-adjusted)
Salaried Default Rate	34.6% (lowest employment segment)
Informal Default Rate	55.0% (highest employment segment)
Agent Channel Default Rate	47.7% (worst channel)
Referral Channel Default Rate	39.2% (best channel — 8.5 pp gap)
NTC Customers	3,488 (34.88% of portfolio)
#1 SHAP Predictor	Interest Rate
#2 SHAP Predictor	DTI
Projected Post-Policy Default	~26–28% (-14 to -16 pp reduction)
Volume Retained Post-Policy	75–80% of current loan volume

7. Dashboard Instructions

The Dashboard.html file is a self-contained interactive dashboard. It requires no server, no internet connection, and no installation.

Step	Action
1	Locate Dashboard.html in the submission ZIP folder
2	Double-click the file — it will open in your default web browser
3	All charts are interactive — hover to see exact values
4	The dashboard works offline — no internet required
5	Tested on Chrome, Firefox, and Edge

The dashboard contains: (1) Default Rate by Grade bar chart, (2) Default Rate by Channel bar chart, (3) CLV vs Risk scatter by Employment, (4) Expected Revenue by Grade chart, (5) Early Warning KPI table, (6) Portfolio summary statistics.

8. Reproducibility

Full reproducibility guaranteed. The synthetic dataset is generated with `numpy.random.seed(42)` at the top of the notebook. Running the notebook from top to bottom will always produce identical results — same dataset, same model performance, same charts, same numbers as reported in the Credit Policy Report.

All random operations in the notebook use seed 42, including: dataset generation, train/test split (`random_state=42`), XGBoost (`random_state=42`), and SHAP sample selection (`random_state=42`).

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